



11 PHYSICS

MEASUREMENT
& DATA

MEASUREMENT AND DATA

Units and Symbols

For almost all of the work done in Years 11 and 12, the SI (Système International) system of units will be used. There are just seven *fundamental quantities* that form the basis of all measurements in science.

Fundamental quantity	SI unit	SI unit symbol
mass	kilogram	kg
length	metre	m
time	second	s
electric current	ampere	A
temperature	kelvin	K
luminous intensity	candela	cd
amount of substance	mole	mol

A *derived unit* is a combination of fundamental units.

e.g. A derived unit is the unit of velocity: metre per second. It is written $m s^{-1}$.

Quantity	SI unit	SI unit symbol	Equivalent unit
velocity	metres per second	$m s^{-1}$	—
acceleration	metres per second per second	$m s^{-2}$	—
frequency	hertz	Hz	s^{-1}
force	newton	N	$kg m s^{-2}$
energy/work	joule	J	$kg m^2 s^{-2}$

Scientific Notation

Measurements are often written in scientific notation. Quantities are written as a number between one and ten and then multiplied by an appropriate power of ten.

(Note that ‘scientific notation’, ‘standard notation’ and ‘standard form’ all have the same meaning.)

e.g. Some measurements written in scientific notation are:

$$\begin{aligned}0.054 \text{ m} &= 5.4 \times 10^{-2} \text{ m} \\245.7 \text{ J} &= 2.457 \times 10^2 \text{ J} \\2080 \text{ N} &= 2.080 \times 10^3 \text{ N}\end{aligned}$$

You should be routinely using scientific notation to express numbers. An important reason for using scientific notation is that it removes ambiguity about the precision of some measurements.

e.g. A measurement recorded as 240 m could be a measurement to the nearest metre.
 $\therefore$ Use 2.40×10^2 m.

The measurement may be to the nearest 10 m.
 $\therefore$ Use 2.4×10^2 m.

Prefixes and Conversion Factors

You should be familiar with the prefixes and conversion factors in the table below. The table gives all conversions as a ***multiplying factor***.

Simply multiply the given number by the factor.

Multiplying factor		Prefix	Symbol
1 000 000 000 000	10^{12}	tera	T
1 000 000 000	10^9	giga	G
1 000 000	10^6	mega	M
1 000	10^3	kilo	k
0.01	10^{-2}	centi	c
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	μ
0.000 000 001	10^{-9}	nano	n
0.000 000 000 001	10^{-12}	pico	p

e.g. $2.45 \times 10^3 \text{ mm} = 2.45 \times 10^3 \times 10^{-3} = 2.45 \text{ m}$

$2.45 \times 10^3 \text{ nm} = 2.45 \times 10^3 \times 10^{-9} = 2.45 \times 10^{-6} \text{ m}$

$1.25 \times 10^3 \text{ mm}^2 = 1.25 \times 10^3 \times (10^{-3})^2 = 1.25 \times 10^{-3} \text{ m}^2$

When calculating an area conversion, the factor is "squared".
 For a volume conversion, the factor is "cubed".

To convert km h⁻¹ to m s⁻¹, divide by 3.60.

e.g. $70.0 \text{ km h}^{-1} = 70.0 \div 3.60 = 19.4 \text{ m s}^{-1}$

Examples

Worked example B1

The diameter of a cylindrical piece of copper rod was measured at 24.8 mm with a vernier caliper. Its length was measured at 35 cm with a tape measure.	
a	Find the area of cross-section in m^2 .
b	Find the volume of the copper rod in m^3 .
Answer	
a	The area of cross-section is πr^2 . The radius is calculated by dividing the diameter by two. Hence the radius is 12.4 mm. To calculate the area in m^2 , first halve the diameter and convert it to metres. The radius is $\frac{24.8}{2} = 12.4 \text{ mm} = 12.4 \times 10^{-3} \text{ m}$. The radius is not written in scientific notation. This is not necessary. All you need to do is multiply by the appropriate factor. The conversion factor for mm to m is 10^{-3} . Just multiply by the conversion factor and don't bother to rewrite the result in scientific notation. This is because it is only going to be used in a calculation and is not a final result. The area of cross-section is $\pi r^2 = \pi(12.4 \times 10^{-3})^2 = 4.8 \times 10^{-4} \text{ m}^2$.
b	The volume is $\pi r^2 h$, where h is the length of the cylinder. The length is $35 \text{ cm} = 35 \times 10^{-2} \text{ m}$. Hence the volume is $\pi(12.4 \times 10^{-3})^2(35 \times 10^{-2}) = 1.7 \times 10^{-4} \text{ m}^3$.

Worked example B2

a	A car is traveling at 110 km h^{-1} . How fast is this in ms^{-1} ?
b	Convert 35 miles per hour to metres per second. A mile is approximately 1600 m.
Answer	
a	110 km h^{-1} is 110×10^3 metres per 3600s. $\frac{110 \times 10^3}{3600} = 30.6$ Hence $110 \text{ km h}^{-1} = 30.6 \text{ ms}^{-1}$.
b	35 miles per hour is 35×1600 metres per 3600s. $\frac{35 \times 1600}{3600} = 15.6$ Hence $35 \text{ mph} = 16 \text{ m s}^{-1}$.

Uncertainty in Measurements

All measurements have some amount of uncertainty.

The uncertainty is one half of the finest scale division on the measuring instrument.

e.g. A measuring wheel has an uncertainty of 0.5 m (if it measures to the nearest metre).

A metre rule has an uncertainty of 0.5 mm (smallest division is 1.0 mm)

An electronic balance set to measure grams to two decimal places has an uncertainty of 0.005 g.

Sometimes this uncertainty is referred to as error. ***It is not error***, in that it is not a mistake or something wrong. All measuring instruments have limited precision and the uncertainty is half of the smallest scale division on the instrument.

The uncertainty gives us the range in which a measurement falls. If we measured the length of a stick with a metre rule then we would get a measurement 'plus or minus' half a millimetre.

e.g. Any stick between 127.5 and 128.5 mm long would be measured as 128 mm to the nearest millimetre. We would record this as ***$128 \pm 0.5 \text{ mm}$*** .

The best judgement you can definitely claim is one half of a scale division - it is too difficult to judge to a finer degree. The uncertainty we will still assume, however, is a full half scale division.

e.g. You might measure another stick, one that has a length somewhere between 154 and 155 mm, as ***$154.5 \pm 0.5 \text{ mm}$*** .

There are two ways to record uncertainties.

Absolute uncertainty: include the actual uncertainty in the measurement.

e.g. $154.5 \pm 0.5 \text{ mm}$

Percentage uncertainty: expressed as a percentage of the measurement.

e.g. % uncertainty = $0.5/154.5 \times 100\% = 0.3\%$

$\therefore 154.5 \pm 0.3 \%$

Uncertainty in Calculations

An experiment or a measurement exercise is not complete until the uncertainties have been analysed. The report should include an estimate of the total uncertainty. This gives the reader of the report some idea of your confidence in the result.

The following three processes are used for estimating uncertainty.

- When adding or subtracting data, add the absolute uncertainties.
- When multiplying or dividing data, add the percentage uncertainties.
- When raising data to power n , multiply the percentage uncertainty by n .

e.g. Calculate the area of a rectangle measuring 22.6 ± 0.1 mm ($\pm 0.4\%$) by 14.2 ± 0.1 mm ($\pm 0.7\%$)

$$A = (22.6)(14.2) = 3.91 \times 10^2 \text{ mm}^2 \pm 1.1\% \\ = \underline{3.91 \times 10^2 \pm 4 \text{ mm}^2}$$

Example

You might have measured the specific heat capacity of a metal. You could have calculated your result using:

$$c_{\text{metal}} = \frac{c_{\text{water}} m_{\text{water}} \Delta T_{\text{water}}}{m_{\text{metal}} \Delta T_{\text{metal}}}$$

Suppose you had the following data included in your table.

Quantity		Absolute uncertainty	Percentage uncertainty
c_{water}	$4180 \text{ J kg}^{-1} \text{ K}^{-1}$	$5 \text{ J kg}^{-1} \text{ K}^{-1}$	0.120
m_{water}	$72.5 \times 10^{-3} \text{ kg}$	$0.05 \times 10^{-3} \text{ kg}$	0.069
ΔT_{water}	5°C	1°C^*	20
m_{metal}	$87.3 \times 10^{-3} \text{ kg}$	$0.05 \times 10^{-3} \text{ kg}$	0.057
ΔT_{metal}	72°C	1°C^*	1.389

*Note that the ΔT values have an absolute uncertainty of 1°C because they are calculated by subtracting one temperature measurement from another.

You would calculate as follows:

$$c_{\text{metal}} = 241 \text{ J kg}^{-1} \text{ K}^{-1} \\ \text{Uncertainty (\%)} = 0.120 + 0.069 + 20 + 0.057 + 1.389 \\ = 21.6\%$$

Hence, you would obtain the following result:

$$c_{\text{metal}} = 241 \text{ J kg}^{-1} \text{ K}^{-1} \pm 21.6\% \\ c_{\text{metal}} = 241 \pm 52 \text{ J kg}^{-1} \text{ K}^{-1}$$

Once you have done all of this you can consider the relative success of your measurement exercise.

Your result is:

$$189 \text{ J kg}^{-1} \text{ K}^{-1} \leq c_{\text{metal}} \leq 293 \text{ J kg}^{-1} \text{ K}^{-1}$$

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In some classes, students are instructed to quote all results to two decimal places or to three significant figures. You should be able to see from Worked example B3 that these rules are not absolutely correct when applied to real data. For ordinary calculations in assignments, tests and examinations, you might just give your answers to three figures.

If a calculation is done in several stages then you should not round off any intermediate results. This will add rounding error to your calculations. Use the memory on your calculator so that there is no rounding until the end of your calculation.

Significant Figures

Significant figures represent a way of expressing how confident we are in a measurement. The final digit carries an element of uncertainty - it has been rounded off.

e.g. An electronic balance measures a mass as 143.24 g.

This has 5 significant figures. The machine rounds the second decimal off to the nearest number - it is the uncertain number.

Rules To Use

- All non-zero digits are significant.
- Zeroes between non-zero digits are significant.
e.g. 4003 has 3 s.f.
- Zeroes to the right of a decimal point and following a non-zero digit are significant.
e.g. 1.102×10^3 has 4 s.f. 0.0105 has 3 s.f.
- All other zeroes are not significant.
e.g. 500 has 1 s.f.

If the zeroes are significant, write it in scientific notation - 5.00×10^2 has 3 s.f.

- In calculations, the final result should not contain any more significant figures than the data with the least number of significant figures.

e.g. $2.45 \times 5.6 \times 103.23 = 1.4 \times 10^3$

5.6 has 2 s.f. so the answer must have 2 s.f. - it has the least number of s.f.

GRAPHICAL ANALYSIS OF DATA

Scientists like to take a large amount of data and observe the trends in that data. This gives more precise measurements and allows scientists to recognise and eliminate problematic data. Physicists commonly use graphical techniques to analyse a set of data.

Linear Relationships

Linear relationships and their graphs are fully specified with just two numbers: gradient, m , and vertical axis intercept, c . In general, linear relationships are written:

$$y = mx + c$$

The gradient, m , can be calculated from the coordinates of two points on the line.

$$m = \frac{\text{rise}}{\text{run}} \\ = \frac{y_2 - y_1}{x_2 - x_1}$$

where (x_1, y_1) and (x_2, y_2) are any two points on the line.

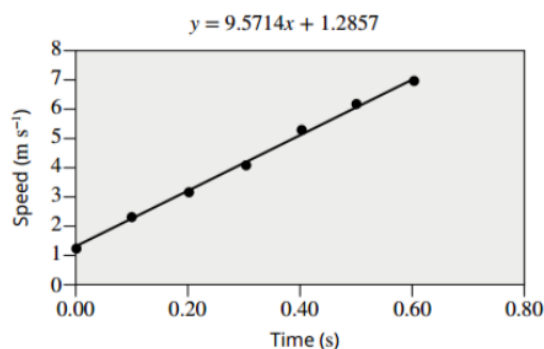
Don't forget that m and c have units. Omitting these is a common error.

Example

Some students used a motion detector to analyse the motion of a falling tennis ball. The relevant relationship is $v = u + at$, where v is the speed of the ball at any given time, u is the initial speed when measurements began, and a is the acceleration of the ball. Below is the collected data.

Time (s)	Speed (m s ⁻¹)
0.0	1.25
0.1	2.30
0.2	3.15
0.3	4.10
0.4	5.25
0.5	6.10
0.6	6.95

Determine the acceleration graphically.



The data is assumed linear with the relationship $v = u + at$, which can be thought of as being $v = at + u$, which makes it clear that putting v on the vertical axis and t on the horizontal axis gives a linear graph with gradient a and vertical intercept u .

i.e. $v \propto t$

The regression line as $y = 9.57x + 1.2857$. If this is rearranged and the constants are suitably rounded, the equation is $v = 1.3 + 9.6t$.

This indicates that the ball was moving at 1.3 ms^{-1} at the commencement of data collection and the ball was accelerating at 9.6 ms^{-2} .

Manipulating Non-Linear Data

Suppose we were examining the relationship between two quantities B and d and had good reason to believe that the relationship between them is

$$B = \frac{k}{d}$$

where k is some constant value.

$$\text{i.e. } B \propto \frac{1}{d}$$

Hence plotting B against $\frac{1}{d}$ will produce a linear graph, with the gradient equal to the constant k.

Example

Some students were investigating the relationship between current and resistance for a new solid-state electronic device. They obtained the data shown in the table.

Current, I (A)	Resistance, R (Ω)
1.5	22
1.7	39
2.2	46
2.6	70
3.1	110
3.4	145
3.9	212
4.2	236

The students believed that the relationship between i and R is:

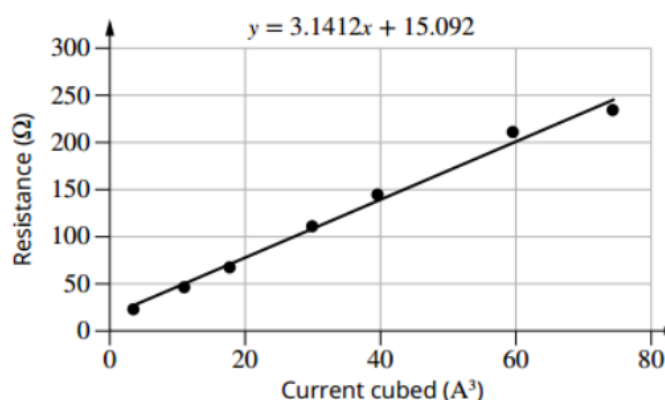
$$R = di^3 + g$$

where d and g are constants.

$$\text{i.e. } R \propto i^3$$

$\therefore$ Plot R against i^3 .

Current cubed, I^3 (A ³)	Resistance, R (Ω)
3.38	22
10.65	46
17.58	70
29.79	110
39.30	145
59.32	212
74.09	236



The regression line has the equation $y = 3.1x + 15.1$.

$\therefore$ The equation relating i and R is $R = 3.1i^3 + 15.1$.

Hence, the value of d is $3.1 \Omega A^{-3}$ and the value of g is 15.1Ω .

Note

In all graphs, it is possible to **plot the error bounds** associated with each piece of data. The line of regression (or "line of best fit") should go through these error bounds.

If a point does not do this, it is considered an "outlier" and can be removed from consideration when determining the line of regression.

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Graphs

When analysing data from a linear relationship, it is first necessary to obtain a graph of the data and an equation for the line that best fits the data. This line of best fit is often called the trendline or the regression line. The entire process can be done on paper but most people will use a computer spreadsheet, the capabilities of a scientific or a graphics calculator, or some other computer-based process. In what follows, it is not assumed that you are using any particular technology.

If you are plotting your graph manually on paper then proceed as follows:

- 1 Plot each data point on clearly labelled, unbroken axes.
- 2 Identify and label but otherwise ignore any suspect data points.
- 3 Draw, by eye, the 'line of best fit' for the points. The points should be evenly scattered either side of the line.
- 4 Locate the vertical axis intercept and record its value as ' c '.
- 5 Choose two points on the line of best fit to calculate the gradient. Do not use two of the original data points as this will not give you the gradient of the line of best fit.
- 6 Write $y = mx + c$, replacing x and y with appropriate symbols, and use this equation for any further analysis.

If you are using a computer or a graphics calculator then proceed as follows:

- 1 Plot each data point on clearly labelled, unbroken axes.
- 2 Identify suspect data points and create another data table without the suspect data.
- 3 Plot a new graph without the suspect data. Keep both graphs, as you don't actually discard the suspect data, but do eliminate it from the analysis.
- 4 Plot the line of best fit—the regression line. The manner in which you do this depends on the model of calculator or the software being used.
- 5 Compute the equation of the line of best fit that will give you values for m and c .
- 6 Write $y = mx + c$, replacing x and y with appropriate symbols, and use this equation for any further analysis.